

Advances in Retrieval-Augmented Generation for Enterprise Document Intelligence

Abstract

Retrieval-Augmented Generation (RAG) has emerged as a dominant paradigm for grounding large language model (LLM) outputs in verified external knowledge. This paper surveys recent architectural improvements including hybrid dense-sparse retrieval, reranking strategies, and agentic multi-step reasoning chains. We evaluate these on a novel enterprise benchmark comprising 10,000 heterogeneous documents across six domain verticals and report RAGAS scores alongside human preference ratings. Our best system achieves a faithfulness score of 0.91 and an answer relevancy of 0.87.

1. Introduction

Enterprise knowledge management suffers from information fragmentation: critical facts are buried in PDFs, spreadsheets, internal wikis, and email threads. LLMs alone hallucinate when queried beyond their training distribution. RAG bridges this gap by decomposing a query into a retrieval step and a generation step, conditioning the model on retrieved passages at inference time (Lewis et al., 2020).

2. Related Work

Early RAG implementations relied on BM25 sparse retrieval. DPR (Karpukhin et al., 2020) introduced dense passage retrieval with bi-encoder architectures, achieving significant gains on open-domain QA. Subsequent work (Izacard & Grave, 2021) demonstrated that fusion-in-decoder approaches outperform single-passage generation by attending over multiple retrieved documents simultaneously.

3. Architecture

Our system comprises four stages: (1) document ingestion with OCR fallback for scanned pages, (2) semantic chunking with 500-character windows and 50-character overlap, (3) hybrid retrieval combining cosine similarity on MiniLM-L6 embeddings with BM25 keyword matching, and (4) LLaMA-3 70B generation with citation-aware prompting. ChromaDB provides persistent vector storage with HNSW indexing for sub-100ms retrieval at 1M+ chunk scale.

4. Results

Model	Faithfulness	Answer Relevancy	Context Recall
BM25 + GPT-3.5	0.71	0.74	0.68
DPR + GPT-4	0.83	0.81	0.79
Hybrid + LLaMA-3 70B (ours)	0.91	0.87	0.88

Table 1: RAGAS evaluation on the enterprise benchmark.

5. Conclusion

Hybrid retrieval combined with citation-aware generation substantially improves factual grounding in enterprise RAG systems. Future work will explore multi-hop reasoning and dynamic knowledge graph integration.